

The performance of VideoChat-T on MVBench is shown in Table 9. Compared to VideoChat2, VideoChat-T only experienced a 0.5% accuracy loss. This indicates that our method effectively preserves the capabilities of the base model, preventing catastrophic forgetting caused by incremental fine-tuning. For a detailed analysis of the performance degradation of MVBench, please refer to Appendix F.2. For the Action Localization (AL) task, which requires the model to determine the coarse-grained temporal position of events, the test accuracy improved from 44.0% to 56.5%. This indirectly confirms that our method significantly enhances the model’s temporal awareness capabilities.

Model	LLM size	Overall (%)		Short Video (%)		Medium Video (%)		Long Video (%)	
		w/o subs	w subs	w/o subs	w subs	w/o subs	w subs	w/o subs	w subs
ST-LLM (Liu et al., 2024b)	7B	37.9	42.3	45.7	48.4	36.8	41.4	31.3	36.9
Video-LLaVA (Lin et al., 2023a)	7B	39.9	41.6	45.3	46.1	38.0	40.7	36.2	38.1
ShareGPT4Video (Chen et al., 2024)	8B	39.9	43.6	48.3	53.6	36.3	39.3	35.0	37.9
Chat-UniVi-v1.5 (Jin et al., 2024)	7B	40.6	45.9	45.7	51.2	40.3	44.6	35.8	41.8
Qwen-VL-Chat (Bai et al., 2023)	7B	41.1	41.9	46.9	47.3	38.7	40.4	37.8	37.9
ShareGemini (Share, 2024)	7B	43.2	47.9	49.1	52.8	41.3	47.3	39.1	43.4
VideoChat2 (Li et al., 2024b)	7B	39.5	43.8	48.3	52.8	37.0	39.4	33.2	39.2
VideoChat-T	7B	46.3	55.8	53.3	59.9	43.8	54.0	41.9	53.4

Table 10: The full performance of VideoChat-T on VideoMME. VideoChat-T achieved significant performance improvements, particularly in the long video subset.

The overall performance of our model on VideoMME is presented in Table 10. VideoChat-T achieved significant improvements on both evaluation benchmarks of VideoMME, which include watching videos only and videos with subtitles. The improvements are particularly notable in the long video subset.

F EXTRA ABLATION

F.1 DOMAIN CORRELATION OF DATA

Model	Charades-STA(R@1 IOU=0.5)	MVBench(avg)
VideoChat-T	48.7	59.9
w/o STAR	47.5 (-1.2)	59.4 (-0.5)

Table 11: The performance changes of the model after removing STAR. Although the video sources of STAR may have some domain correlation with those of Charades-STA and MVBench, the performance of our model is minimally affected by STAR.

We found that the video sources in the STAR dataset might have some domain correlation with the video sources in MVBench and Charades-STA. Therefore, we removed STAR from the training set while keeping other training settings consistent with the original. The performance on benchmarks where the video sources might have domain correlation is shown in Table 11. The model’s accuracy on Charades-STA (R@1 IOU=0.5) decreased by 1.2%, and the average accuracy on MVBench decreased by 0.5%. This indicates that the domain correlation of video sources did not significantly impact performance for our model. Notably, after removing STAR, our normal data volume was reduced to approximately 36K. This implies that, with sufficiently parameter-efficient initialization and appropriate training strategies, using only a small amount of high-quality normal data is sufficient to retain the model’s original capabilities.

F.2 DEEPER INVESTIGATION OF THE PERFORMANCE DROP ON MVBENCH

We conducted a deeper investigation into the performance decline on MVBench. Through additional ablation experiments (as shown in Table 12), we identified two main factors contributing to the performance drop.

Architectural Discrepancy: The original VideoChat2 model was designed to process only 16 frames, leading to a mismatch in the learned feature distribution compared to the architecture of VideoChat-T. As shown in the first two rows of the table, increasing the input frame number for VideoChat2

Method	post ft data	data size	frame num	token num (per frame)	MVBench(AVG)
VideoChat2	-	-	16	12	60.4
VideoChat2	-	-	128	12	42.1
VideoChat-T (Common_Init)	-	-	128	3	25.3
VideoChat-T (Ours)	-	-	128	3	48.6
VideoChat-T (Ours)	TimePro+Normal (Ours)	0.43M	128	3	59.9
VideoChat-T (Ours)	TimePro+FullVideoChat2	2M	128	3	62.9

Table 12: Performance of VideoChat2 and VideoChat-T on MVBench under different settings.

resulted in a significant performance drop (from 60.4 to 42.1). When initializing VideoChat-T with VideoChat2, performance was close to random (25.3) due to the newly introduced randomly initialized layers. By applying efficient initialization to these new layers, we partially recovered the original capabilities of the model, bringing the MVBench performance of the un-trained VideoChat-T back to 48.6, representing an improvement of 6.5 compared to the 128-frame VideoChat2. After further fine-tuning, the short-video processing capability of VideoChat-T improved significantly, reaching 59.9.

Fine-tuning Data Discrepancy: We fine-tuned VideoChat-T using only 432K data, significantly less than the 2M non-grounded regular data used for training VideoChat2. The fine-tuning data for VideoChat2 primarily consisted of short videos of around ten seconds, which closely matched the length distribution of the MVBench evaluation videos, playing a crucial role in improving MVBench performance. To validate our hypothesis, we conducted additional experiments by training our VideoChat-T model using the TimePro and full VideoChat2 training data. It can be observed that VideoChat-T showed a slight improvement in performance on the MVBench dataset, achieving an accuracy of 62.9, which is an increase of 2.5 compared to the original VideoChat2.

Based on the above, we can conclude the fundamental reasons affecting the model’s foundational generalization capabilities. When a model undergoes adjustments, the learned original distribution may not perfectly match the new architecture, making the efficient initialization of new layers crucial. The features learned from the original dataset might be forgotten due to changes in various parameters. Utilizing a more comprehensive and diverse dataset for fine-tuning can restore and even further enhance performance.

F.3 ASSOCIATION BETWEEN PERFORMANCE AND MODEL DESIGN

Method	FT Data	Charades-STA	QVHighlight	MVBench	Egoschema	VideoMME
		IOU0.5	mAP	Avg	Full	w/o subs
TimeChat	TimeIT+Valley	32.2	14.5	38.5	33.0	30.2
TimeChat	TimePro+Normal	34.2	16.3	41.6	38.9	33.4
VideoChat-T	TimePro+Normal	48.7	26.5	59.9	60.0	46.3

Table 13: Comparison of other model architectures trained on our dataset with our method, demonstrating the impact of the overall model structure design.

To eliminate the influence of training data and auxiliary tasks, and to more clearly evaluate the association between performance and model design, we fine-tuned TimeChat using the full set of fine-tuning data and auxiliary tasks from VideoChat-T. Table 13 presents the performance of TimeChat, fine-tuned with our data, across five datasets. It can be observed that TimeChat, fine-tuned with our data, shows improvements across all benchmarks. However, its performance still lags significantly behind VideoChat-T. This indicates that an efficient fine-tuning architecture design and high-quality, diverse datasets are both essential and complementary.

F.4 VALIDATION OF TRANSFERABILITY

To verify the robustness of our TimeSuite for other MLLMs, we transferred our method to Llava-OneVision (Li et al., 2024a). Table 14 shows the performance changes of Llava-OneVision after applying our TimeSuite. It can be seen that when we apply the full set of methods in TimeSuite to Llava-OneVision, the model’s performance on two different long-video evaluation benchmarks

Method	Charades-STA	QVHighlight	VideoMME	MLVU	MVBench
	IOU0.5	mAP	w/o subs	Avg	Avg
Llava-OneVision (baseline)	7.3	15.0	58.2	64.7	56.7
Llava-OneVision-T (Ours)	42.5	21.7	61.4	69.4	56.1

Table 14: Performance comparison of TimeSuite migration to other MLLMs. The application of our method shows a certain improvement in long video comprehension, demonstrating the transferability of our approach.

improves (+3.2 on VideoMME and +4.7 on MLVU), effectively demonstrating the robustness of our TimeSuite for different MLLMs.

F.5 EXPLORATIONS OF DATA CONFIGURATIONS OF TIMEPRO

Method	MVBench	Egoschema	VideoMME	Charades-STA	QVHighlight
	Avg	Full	w/o subs	IOU=0.5	mAP
TimePro615K+Normal82K (old version)	60.0	61.0	46.3	45.4	25.7
TimePro349K+Normal82K (Ours)	59.9	60.0	46.3	48.7	26.5

Table 15: Comparison of different versions of our proposed TimePro. More data does not necessarily lead to higher overall performance, highlighting the importance of data quality.

In the early version of TimePro, we employed datasets comprising 309K Multi-format Temporal Grounding instances, 150K Temporal Grounded Caption instances and other data. Through extensive experimentation (as shown in Tabel 15), we discovered that removing low-quality data while retaining high-quality instances could significantly reduce training time without compromising performance. Consequently, we pruned these two part datasets to 100K and 93K instances, respectively. The data distribution presented in the paper represents the optimized and relatively balanced configuration we arrived at.

G DISCUSSION

G.1 CAN THE OVERALL PERFORMANCE OF MLLMs BE ENHANCED BY CONTINUOUSLY INTEGRATING EXPERT TASKS?

By appropriately fine-tuning the Multimodal Large Language Model (MLLM), we have developed a general MLLM with powerful zero-shot temporal grounding capabilities. Its performance, after fine-tuning on the training set of evaluation benchmarks, can rival the current state-of-the-art supervised expert models. Based on these results, we can boldly speculate whether it is possible to internalize the capabilities of expert models such as spatial grounding, tracking and detection (Zeng et al., 2023) into the MLLM itself, without using any external expert decoders, to enhance the comprehensive understanding performance of the MLLM and achieve a unified generalist MLLM for multiple tasks.

Merlin (Yu et al., 2023) and VisionLLM (Wang et al., 2024b) have already attempted something similar, but its performance is limited by the reasoning capabilities and language representation bottlenecks of the LLM. There is still a significant gap between its performance and that of expert models for various tasks. We observed similar phenomena in our experiments. The temporal grounding task only requires outputting two timestamps, and the task format is relatively simple, so our model achieved good results. However, the highlight detection task requires outputting multiple discrete timestamps and their corresponding saliency scores. The model needs to accurately predict dozens of numbers in language form to answer the question correctly. Our model performed well only on data with fewer timestamps. Therefore, how to simplify the complex output format of expert tasks into the language representation of LLMs, or to design special processing procedures to simplify complex expert tasks, is a question worth exploring.

Moreover, designing diverse data formats is also crucial for enhancing the expert capabilities of MLLMs. Compared to classic expert models, MLLMs have a natural advantage in task type diversity and can enhance their performance through various different variants tasks of a single capability.